

99.500 Applied Mathematics for Engineering Spring Semester AY2026

Matlab for PDE

Week 12 Use Matlab to solve PDE

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≡ CONTENTS

[« Documentation Home](#)

[« MATLAB](#)

[« Mathematics](#)

[« Numerical Integration and Differential Equations](#)

[« 1-D Partial Differential Equations](#)

pdepe

ON THIS PAGE

[Syntax](#)

[Description](#)

[Examples](#)

[Input Arguments](#)

[Output Arguments](#)

[Tips](#)

[Algorithms](#)

Documentation

Examples

Functions

Apps

Videos

Answers

pdepe

Solve 1-D parabolic and elliptic PDEs [colla](#)

Syntax

```
sol = pdepe(m,pdefun,icfun,bcfun,xmesh,tspan)
sol = pdepe(m,pdefun,icfun,bcfun,xmesh,tspan,options)
[sol,tso1,sole,te,ie] = pdepe(m,pdefun,icfun,bcfun,xmesh,tspan,options)
```

Description

`sol = pdepe(m,pdefun,icfun,bcfun,xmesh,tspan)` solves a system of parabolic and elliptic PDEs with one spatial variable x and time t . At least one equation must be parabolic. The scalar `m` represents the symmetry of the problem (slab, cylindrical, or spherical). The equations being solved are coded in `pdefun`, the initial value is coded in `icfun`, and the boundary conditions are coded in `bcfun`. The ordinary differential equations (ODEs) resulting from discretization in space are integrated to obtain approximate solutions at the times specified in `tspan`. The `pdepe` function returns values of the solution on a mesh provided in `xmesh`.

Syntax for “pdepe”

```
sol = pdepe(m,pdefun,icfun,bcfun,xmesh,tspan)
```

$$c\left(x,t,u,\frac{\partial u}{\partial x}\right)\frac{\partial u}{\partial t} = x^{-m}\frac{\partial}{\partial x}\left[x^m f\left(x,t,u,\frac{\partial u}{\partial x}\right)\right] + s\left(x,t,u,\frac{\partial u}{\partial x}\right)$$

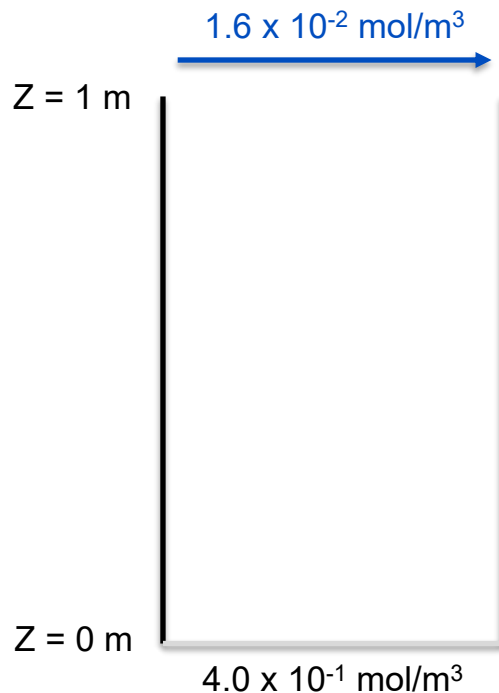
```
[c,f,s] = pdefun(x,t,u,dudx)
```

with $\begin{cases} IC : & u(x,t_0) = u_0(x) \\ BC1 : & pl(x_l,t,u_l) + ql(x_l,t)f\left(x_l,t,u_l,\frac{\partial u}{\partial x}\Big|_l\right) = 0 \\ BC2 : & pr(x_r,t,u_r) + qr(x_r,t)f\left(x_r,t,u_r,\frac{\partial u}{\partial x}\Big|_r\right) = 0 \end{cases}$

```
function [pl,ql,pr,qr] = bcfun(xl,ul,xr,ur,t)
```

Chimney Example by Tan Xue Feng

We consider a chimney of height 1m that evacuates CO_2 as shown below. Suppose that initially (time $t < 0$), the chimney remains closed such that the concentration of CO_2 gas in the chimney is 0 mol/m^3 . At time $t = 0$, the chimney is suddenly open at the top, where ambient air with a steady CO_2 concentration of $1.6 \times 10^{-2} \text{ mol/m}^3$ flows over. At the same time, flue gas at $4.0 \times 10^{-1} \text{ mol/m}^3$ is introduced at the bottom of the chimney. We assume that ambient air above does not enter the chimney, and the primary mode of CO_2 through the chimney is through diffusion.



Governing equation

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial z^2}$$

where D is the diffusion coefficient

Boundary conditions

$$c(0, t) = 4.0 \times 10^{-1}$$

$$c(1, t) = 1.6 \times 10^{-2}$$

Initial conditions

$$c(z, 0) = 0$$

Rearrange PDE into Standard Form

Standard form:

$$c \left(x, t, u, \frac{\partial u}{\partial x} \right) \frac{\partial u}{\partial t} = x^{-m} \frac{\partial}{\partial x} \left(x^m f \left(x, t, u, \frac{\partial u}{\partial x} \right) \right) + s \left(x, t, u, \frac{\partial u}{\partial x} \right)$$

Rearranging equation into standard form:

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial z^2} \longrightarrow 1 \frac{\partial c}{\partial t} = \frac{\partial}{\partial z} \left[D \frac{\partial c}{\partial z} \right] + 0 \quad \left| \quad \begin{array}{l} c = 1 \\ f = D \frac{\partial u}{\partial z} \\ s = 0 \\ m = 0 \end{array} \right.$$

Define function that represent the PDE function:

```
function [c,f,s] = pdefun (z,t,u,DuDz)
    D = 1e-3;
    c = 1;
    f = D*DuDz;
    s = 0;
end
```

Rearrange BCs into Standard Form

Standard form:

$$p(x, t, u) + q(x, t) f \left(x, t, u, \frac{\partial u}{\partial x} \right) = 0$$

Rearranging BCs into standard form:

$$\begin{array}{lcl} c(0, t) = 4.0 \times 10^{-1} \longrightarrow (ul - 0.4) + 0 & \left| \right. & pl = ul - 0.4 \\ & & ql = 0 \\ c(1, t) = 1.6 \times 10^{-2} \longrightarrow (ur - 0.016) + 0 & \left| \right. & pr = ur - 0.016 \\ & & qr = 0 \end{array}$$

Define function that represent the boundary condition:

```
function [pl,ql,pr,qr] = bcfun (xl,ul,xr,ur,t)
    pl=ul-0.4;
    ql=0;
    pr=ur-0.016;
    qr=0;
end
```

Define Function for IC, Mesh for z and t

Define a function for initial condition:

```
function [u0] = icfun(x)  
    u0=0;  
end
```

Define a mesh for z and t :

```
z = linspace(0, 1, 1e3);  
t = linspace(0, 1e3, 1e3);
```

Apply PDEPE solver:

```
m=0;  
sol = pdepe(m, @pdefun, @icfun, @bcfun, z, t);
```

Extract results:

```
u = sol
```

Solving PDE with *pdepe* to Obtain 3D Plot

```
% Parameters
z = linspace(0, 1, 1e3); % discretize space
t = linspace(0, 1e3, 1e3); % discretize time
m = 0;

% Solve PDE
sol = pdepe(m, @pdefun, @icfun, @bcfun, z, t);

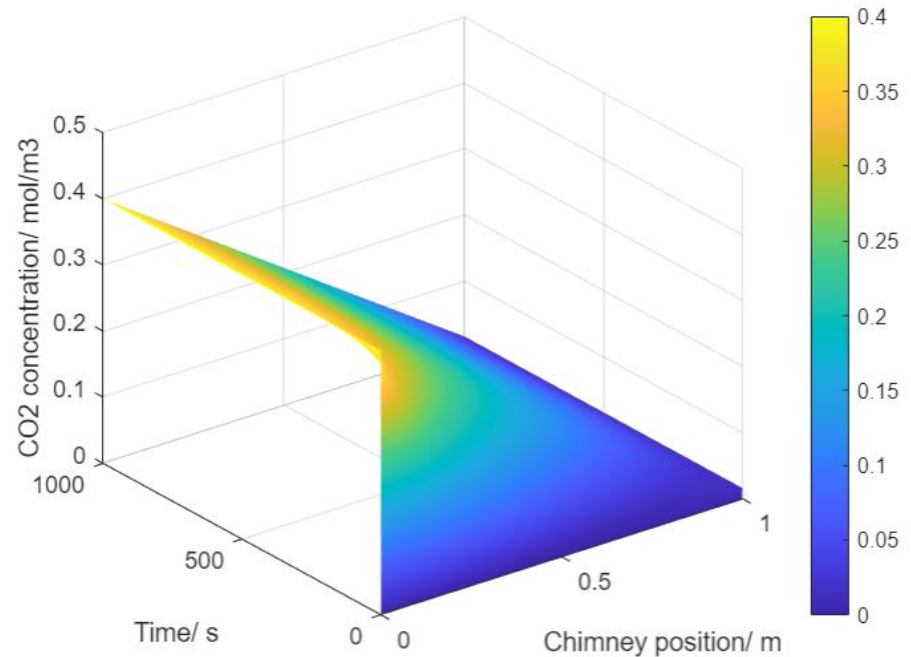
% Extract solution
u = sol;

% Plot solution
surf(z,t,u)
xlabel('Chimney position/ m'); % Label for x-axis
ylabel('Time/ s'); % Label for y-axis
zlabel('CO2 concentration/ mol/m3'); % Label for z-axis
shading interp
colorbar
```

```
function [c,f,s] = pdefun(z,t,u,DuDz)
    D = 1e-3; % Define diffusion coefficient
    c = 1;
    f = D*DuDz;
    s = 0;
end

function [pl,ql,pr,qr] = bcfun(xl,ul,xr,ur,t)
    pl = ul - 0.4;
    ql = 0;
    pr = ur - 0.0016;
    qr = 0;
end

function [u0] = icfun(x)
    u0 = 0;
end
```

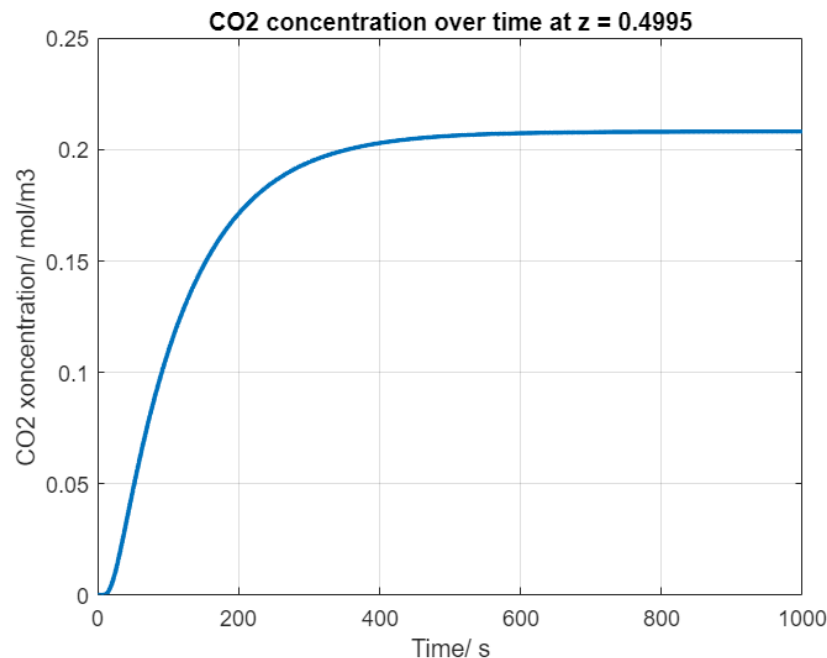


Solving PDE with *pdepe* to Obtain 3D Plot

```
% Find the closest point to z = 0.5
desired_z = 0.5; % Desired spatial position
 [~, z_idx] = min(abs(z - desired_z)); % Index of closest point

% Extract the solution at the closest z
z_sol = sol(:, z_idx); % Extract solution for all time at closest z

% Plot the solution over time
plot(t, z_sol, 'LineWidth', 2);
xlabel('Time/ s');
ylabel('CO2 concentration/ mol/m3');
title(['CO2 concentration over time at z = ', num2str(z(z_idx))]);
grid on;
```



Please solve the equation analytically and compare the analytical solution with the numerical one

Symmetry Constant - m



m – Symmetry constant

0 | 1 | 2

Symmetry constant, specified as one of the values in this table. m specifies the type of problem in the equations specified by `pdefun`. Once read the value of m from the equation.

Value	Description
0	1-D Cartesian coordinates with no symmetry
1	2-D Cylindrical coordinates with azimuthal angular symmetry
2	3-D Spherical coordinates with azimuthal and zenith angular symmetries

Week 11 Heat Conduction Example 1

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial y^2}$$

$$T(y, 0) = T_0$$

$$T(\delta, t) = T_\infty$$

$$\left. \frac{\partial T}{\partial y} \right|_{y=0} = 0$$

What are c, m, f and s?

What are pl, ql, pr and qr?

What is u_0 ?

$$c \left(x, t, u, \frac{\partial u}{\partial x} \right) \frac{\partial u}{\partial t} = x^{-m} \frac{\partial}{\partial x} \left[x^m f \left(x, t, u, \frac{\partial u}{\partial x} \right) \right] + s \left(x, t, u, \frac{\partial u}{\partial x} \right)$$

$$\text{with } \begin{cases} IC : & u(x, t_0) = u_0(x) \\ BC1 : & pl(x_l, t, u_l) + ql(x_l, t) f \left(x_l, t, u_l, \left. \frac{\partial u}{\partial x} \right|_l \right) = 0 \\ BC2 : & pr(x_r, t, u_r) + qr(x_r, t) f \left(x_r, t, u_r, \left. \frac{\partial u}{\partial x} \right|_r \right) = 0 \end{cases}$$

Heat Conduction in Sphere Example

$$\frac{\partial T}{\partial t} = \frac{\alpha}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) + \beta + \gamma T^2$$

$$T(r, 0) = T_0 + \frac{r}{r_0} (T_1 - T_0)$$

$$-k \frac{\partial T}{\partial r} \Big|_{r=r_0} = \delta [T(r_0, t) - T_1]$$

$$\frac{\partial T}{\partial r} \Big|_{r=0} = 0$$

What are c, m, f and s?

What are pl, ql, pr and qr?

What is u_0 ?

$$c \left(x, t, u, \frac{\partial u}{\partial x} \right) \frac{\partial u}{\partial t} = x^{-m} \frac{\partial}{\partial x} \left[x^m f \left(x, t, u, \frac{\partial u}{\partial x} \right) \right] + s \left(x, t, u, \frac{\partial u}{\partial x} \right)$$

$$\text{with } \begin{cases} IC : & u(x, t_0) = u_0(x) \\ BC1 : & pl(x_l, t, u_l) + ql(x_l, t) f \left(x_l, t, u_l, \frac{\partial u}{\partial x} \Big|_l \right) = 0 \\ BC2 : & pr(x_r, t, u_r) + qr(x_r, t) f \left(x_r, t, u_r, \frac{\partial u}{\partial x} \Big|_r \right) = 0 \end{cases}$$

Announcements

Week 13 Computer Test: 22 April 2026 Wed 2:15PM – 4:15PM

- 100 marks rescaled to 20%
- Internet accessible, open-book and bring your own laptop
- Work in pairs and no communication between groups

Group 1	Yuchun Wang	Khong Chee Houe
Group 2	Yim Tat Yuen Bernard	Sergei Reva
Group 3	Wang Danqi	Kangyufan
Group 4	Chun Yong Seow	Soe Sandar Thaw
Group 5	Mundo Triscia Katerina Galvan	Liu Lizhuo
Group 6	Kambakhsh Eskandari	Ronit Dutta
Group 7	Ryan Yeo Wei Jie	Kenneth Koh
Group 8	Natalie Yen Gabriel	Teo Zi Yi
Group 9	Ye Xinyu	Darryl Yap Jun Wei

Announcements

Week 14 Final Exam: 29 April 2026 Wed 2:00PM - 4:00PM

- Think Tank 22 (2.311)
- Closed-book. **two** A-4 cheatsheets for PG and **four** A-4 cheatsheets for UG

Week 14 Individual Meetings

- Date: Tuesday, April 28, 2026
- Time: 10:00 AM – 4:45 PM
- Action Required: Please indicate your time slot using this link

<https://docs.google.com/spreadsheets/d/10sE6iWVvFoSE9Oo5DYvNaOhwu1nqeNufKXq8NkKbHRs/edit?gid=0#gid=0>